

DEXTER FERNANDES

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MOTIVATIONAL LETTER

CBOOST · DEN DOLDER, NL

Application: Experienced AI Vision Engineer

I am applying for the Experienced AI Vision Engineer position at Cboost in Den Dolder. I am a computer vision engineer with an MSc in Robotics from the University of Bristol, and my commercial work has been on industrial sites rather than in a research group: 24/7 perception on BP drilling rigs in Oman, and automated inspection and worker safety systems in Texas for John Deere and Austin Industries. Your responsibilities put reliability, maintainability and deployment efficiency at the top rather than model accuracy, and that is where most of my own work has gone.

At Esbaar I owned a real-time vision system on BP drilling rigs end to end, from dataset curation and detector training through edge deployment and on-site validation. Most of the work turned out to be reliability, not accuracy. Containerised services, health checks and safe restarts cut pipeline crashes by 67%. A watchdog with a REST API meant a failed service came back without anyone driving out to the rig, which mattered more on a remote rig than another point of mAP.

Before that I led four engineers at LivNSense. We built YOLOv5 inspection for John Deere engine paintjobs on a conveyor line, catching paint defects, contamination and colour inconsistency. I also ran the multi-camera pipeline on a moving construction vehicle, where I spent a lot of time on camera and ISP settings across RGB and infrared before the models behaved over 24/7 operation. Inference was C++17 on OpenVINO, across Intel Movidius VPUs and Core i7 industrial PCs. Quantisation-aware training and structured pruning halved the model size with accuracy holding, and throughput went from 5 to 15 FPS.

Optics and lighting is where I would be learning from you. I have tuned acquisition and corrected bad images in software. I have never specified a lens or designed a lighting rig, and I would rather write that down than let you find out in the interview.

On logistics: I am happy to relocate to the Utrecht area and to be in the office three days a week. I am not an EU national, so I would need a Dutch work visa.

If you are open to it, I would like a call about how you scope vision projects with customers, and where you want this part of Cboost to be in two years.

Dexter Fernandes

MSc Robotics (Merit), University of Bristol